

A Real-World Evaluation of LLM Medication Safety Reviews in NHS Primary Care

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Abstract

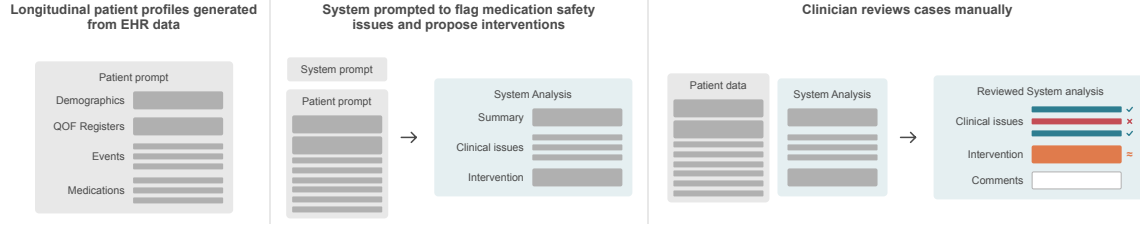
Large language models (LLMs) often match or exceed clinician-level performance on benchmarks, but are rarely evaluated on real clinical data; we present the first real-world evaluation of an LLM medication safety system in NHS primary care. Although issue detection was strong, fully correct issue–intervention pairs occurred in only 46.9% of patients. Errors were driven predominantly by contextual reasoning failures rather than medication knowledge gaps; we provide a categorisation of failures and detailed de-identified vignettes. These findings highlight major barriers to safe clinical deployment.

Large Language Models (LLMs) are rapidly surpassing human expert performance in many domains [1], including aspects of healthcare [2, 3, 4, 5, 6]. This has raised hopes that LLMs could help address the medication safety crisis. The World Health Organization (WHO) estimates medication errors account for approximately 1% of global health expenditure and that preventable medication-related harm (>25% severe or life-threatening) affects around 5% of patients [7, 8, 9]. Medication-related problems represent about half of avoidable medical harm [10], and advanced economies are no exception [11].

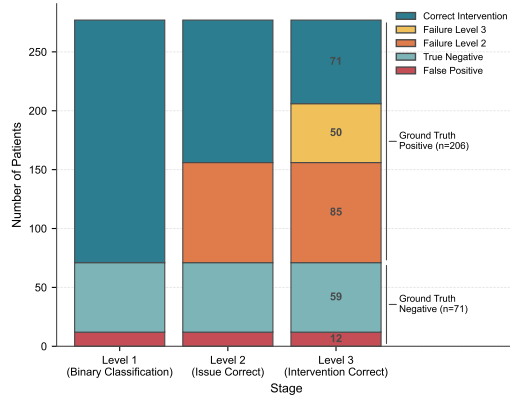
Work in 2025 has shown meaningful progress toward clinician-level medical reasoning in LLMs [3, 2, 6]. Yet, despite accelerating LLM adoption in healthcare [12], recent systematic reviews reveal critical gaps in the evidence base, with only 5% of studies using real

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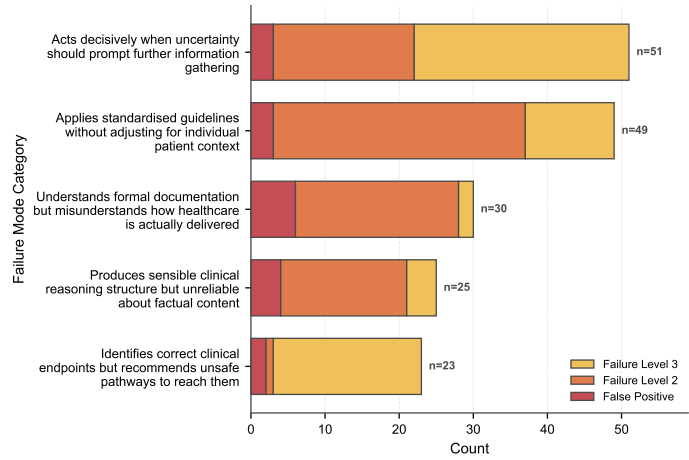
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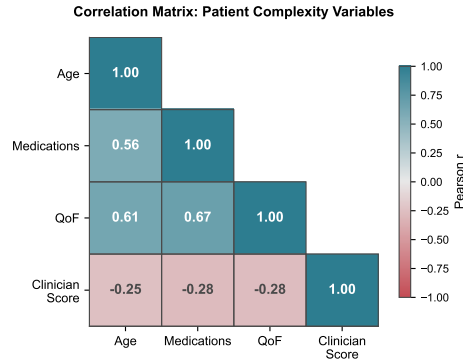
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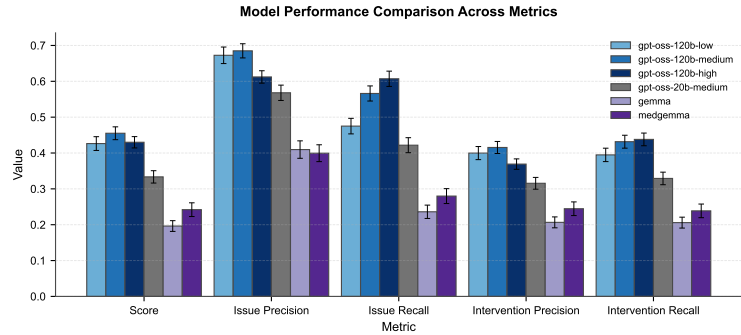


Figure 1: Evaluation framework and primary findings. (a) Evaluation workflow: patient profiles are processed by the System; a clinician reviews outputs marking issues and interventions as correct (✓), partially correct (≈), or incorrect (✗). (b) Hierarchical performance across 277 patients at three stages: binary classification (Level 1), issue correctness (Level 2), and intervention appropriateness (Level 3). (c) Distribution of 178 failure instances by failure reason and evaluation stage. (d) Correlation matrix showing patient complexity variables (age, medication count, QoF registers) and their relationship to clinician scores. (e) Model performance comparison across six configurations.

patient data; the vast majority rely on synthetic patient data or medical-exam-style questions [13, 14], and only a small fraction of these studies report on failure mechanisms [15]. Motivated by this endemic *evaluation gap*, this study assesses state-of-the-art LLMs tasked with conducting medication reviews for complex, real-world primary care cases and provides a detailed characterisation of how these systems fail.

Specifically, we evaluated an LLM-based medication safety system (“System”) using primary care electronic health records for 2,125,549 adults in the NHS Cheshire and Merseyside region, accessed within a Trusted Research Environment (see Appendix G for technical infrastructure and data processing). Structured longitudinal patient profiles were generated from GP events, prescriptions, hospital episodes and observations, and the primary System (gpt-oss-120b [16]) was prompted to flag medication safety issues and propose interventions without external knowledge sources (see Appendix A.3 for full architectural details). A total of 300 patients were selected across a spectrum of clinical complexity using prescribing safety indicators and matched indicator negative controls; System interventions for 277 cases with adequate data were independently reviewed by an experienced clinician, generating a hierarchical score and comprehensive failure characterisation. This is shown in Figure 1a. A complete methodology can be found in Appendix A.

Figure 1b summarises our primary findings. The System showed excellent performance in the initial task of detecting whether a clinical issue was present (accuracy 95.7% [95% CI 92.6–97.5; Wilson score], sensitivity 100% [95% CI 98.2–100], specificity 83.1% [95% CI 72.7–90.1], positive predictive value 94.5% [95% CI 90.6–96.8], negative predictive value 100% [95% CI 93.9–100]). However, it correctly identified all issues and interventions in only 46.9% [95% CI 41.1–52.8] of patients. We observed no cases in which all identified issues were incorrect, and among partially correct cases the precision distribution was left-skewed towards higher values, with a median precision of 60% and only 8.8% of cases below 40%. These findings highlight a gap between detecting that something is wrong and consistently determining what to do about it.

Given the non-blinded evaluation methodology, self-consistency analysis provides an alternative performance ceiling: across repeated runs on identical patient profiles, the System was 87.8% self-consistent at binary classification. See Appendix F for a detailed analysis.

Figure 1d shows that System performance declined with increasing patient-complexity construct captured by age, medication count, and Quality and Outcomes Framework (QoF) condition registration count, each of which correlated negatively with clinician scores ($S_{\text{clinician}} \in [0, 1]$, combining issue identification accuracy (F1) and intervention appropriateness with full/partial/no credit; see Appendix B.1); for example, patients on ≥ 15 medications saw a 22% relative decline in System performance compared to those on 0–4 medications. These three measures were strongly intercorrelated. Counterfactual analysis found no performance variation across White, Asian and Black patients with similar mean scores (White 0.472 ± 0.013 , Asian 0.474 ± 0.013 , Black 0.471 ± 0.013) and no statistically significant variation by calendar date.

Figure 1e shows that performance varied with reasoning effort, model scale, and medical domain fine-tuning: gpt-oss-120b-medium achieved the best overall score (0.459 ± 0.017), outperforming its low- and high-effort counterparts (0.426–0.435) and the smaller gpt-oss-20b-medium (0.334, a 37.4% relative decrease). Gemma-based models performed substantially worse despite similar parameter counts, with medical fine-tuning improving

gemma-3-27b (0.196) to medgemma-27b (0.239) yet still leaving both behind even gpt-oss-20b, suggesting that, for this real-world medication review task, within-architecture scale and general reasoning capacity matter more than domain-specific medical fine-tuning, which may boost performance on synthetic benchmarks but appears insufficient to close the gap in complex real-world cases. Medium reasoning effort also provided the best precision-recall balance (precision 0.689, recall 0.573), with high effort producing more false positives (precision and recall 0.616), consistent with recent work showing that extending test-time reasoning can make models more vulnerable to distractors and spurious patterns, effectively ‘overthinking’ and talking themselves out of initially correct answers [17].

	Overconfidence in uncertainty	Protocol vs patient gap	Protocol vs practice gap	Coherent but factually incorrect	Process blindness
Category	Contextual Reasoning	Contextual Reasoning	Contextual Reasoning	Factual Error	Contextual Reasoning
Description	Acts decisively when uncertainty should prompt further information gathering	Applies standardised guidelines without adjusting for individual patient context, goals of care, or competing risks	Understands formal documentation but misunderstands how healthcare is actually delivered	Produces sensible clinical reasoning structure but is unreliable about specific factual content	Identifies correct clinical endpoints but recommends unsafe pathways to reach them
Failure Modes ● = 1 case	Premature action without information gathering Overlooked clinical issues	Not considering patient specific context Unnecessary flagging Missed deprescription opportunities	Duplicate prescription errors Healthcare system context	Hallucinations Pharmacological knowledge gaps Guideline misapplication	Unsafe medication transitions
Vignette	Vignette 2 HARM Verapamil stopped without confirmed indication PATIENT CONTEXT <i>76 year old woman on long-term warfarin for atrial fibrillation, documented heart failure, and recent verapamil prescription (160mg three times daily)</i> SYSTEM ISSUES ✓ Verapamil contraindicated in heart failure (and may worsen cardiac function) ✓ Supratherapeutic INR on warfarin increasing bleeding risk SYSTEM INTERVENTION ✗ Stop verapamil immediately to remove the contraindicated drug and reduce interaction with warfarin. ✓ Reduce warfarin dose from the current 4.14 mg daily to 3.5 mg daily and repeat INR in 3-4 days. CLINICIAN REVIEW Understanding the original reason for verapamil prescription (rate control in AF) before changing it would be essential (consideration should be given to safer alternatives and discussion with specialist).	Vignette 9 HARM Recommends restarting multiple prescriptions in patient with advanced age and frailty PATIENT CONTEXT <i>91-year-old female with advanced frailty, coronary heart disease, CKD stage 3b, and hypertension. Resides in nursing home with palliative goals of care. Previous statin and ACE-inhibitor discontinued due to tolerability.</i> SYSTEM ISSUES ✗ Absence of high-intensity statin therapy for secondary prevention of coronary heart disease ✗ Absence of ACE-inhibitor therapy for hypertension, CKD protection and cardiovascular risk reduction SYSTEM INTERVENTION ✗ Restart atorvastatin 40 mg once daily. Restart ramipril 2.5 mg once daily. CLINICIAN REVIEW There is nothing technically incorrect about restarting these prescriptions, but they are not clinically appropriate in the context of advanced frailty where the risk would likely outweigh the benefit.	Vignette 22 HARM Doesn't recognise dual prescription achieves intended dose PATIENT CONTEXT <i>58-year-old male with hypertension and type 2 diabetes. Current medications include perindopril 2 mg and perindopril 4 mg (both active prescriptions), metformin 1g twice daily, and amlodipine 5 mg.</i> SYSTEM ISSUES ✗ Duplicate perindopril prescriptions (2 mg and 4 mg tablets) both active, leading to potential dosing confusion and unnecessary complexity. SYSTEM INTERVENTION ✗ Discontinue one of the perindopril prescriptions and consolidate to a single prescription (e.g. prescribe a single 6 mg tablet). CLINICIAN REVIEW Total intended dose is 6 mg. No 6 mg tablet exists; dual prescriptions are the standard approach.	Vignette 27 HARM Hallucinates that Monomil is a clopidogrel PATIENT CONTEXT <i>74-year-old male with stable angina and previous MI (2019). Current medications include aspirin 75 mg, Monomil XL 60 mg (isosorbide mononitrate), bisoprolol 5 mg, and atorvastatin 40 mg.</i> SYSTEM ISSUES ✗ Prolonged dual antiplatelet therapy (aspirin 75 mg + clopidogrel Monomil XL 60 mg) without documented recent indication, increasing bleeding risk. SYSTEM INTERVENTION ✗ Discontinue Monomil XL (clopidogrel) 60 mg daily after confirming no recent indication for dual antiplatelet therapy; continue aspirin 75 mg daily. CLINICIAN REVIEW Monomil XL is isosorbide mononitrate for angina, not clopidogrel. Patient is not on dual antiplatelet therapy.	Vignette 42 HARM Recommends abrupt cessation of amitriptyline without taper PATIENT CONTEXT <i>79-year-old female with ischaemic heart disease, heart failure (EF 35%), and chronic neuropathic pain. Current medications include amitriptyline 10 mg nightly (for pain), furosemide 40 mg, and ramipril 5 mg.</i> SYSTEM ISSUES ✓ Amitriptyline prescribed despite known ischaemic heart disease and heart failure, posing risk of cardiac arrhythmia and anticholinergic side-effects in an elderly patient SYSTEM INTERVENTION ✗ Stop amitriptyline 10 mg nightly immediately. If needed for sleep or neuropathic pain, consider a safer alternative. CLINICIAN REVIEW Stopping amitriptyline requires a gradual taper as it is very difficult to stop abruptly in practice.

Figure 2: **Taxonomy and examples of System failures.** Classification of 178 failure instances across 148 patients into five failure reasons and their corresponding failure modes. Representative vignettes for each failure type showing the clinical scenario, System output, and clinician assessment. Ticks indicate correct identifications; crosses indicate errors. A full set of vignettes can be found in Appendix C, Table 1.

The failure analysis reveals an inversion of typical assumptions about LLM limitations.

Across 148 patients with identified failures, we observed 178 distinct failure instances (Figure 1c; Figure 2). Factual errors—hallucinations, pharmacological knowledge gaps, and guideline misapplication—accounted for only 25 instances (14%). The remaining 153 instances (86%) reflected contextual reasoning failures: overconfidence prompting premature action (n=51), failure to adjust protocols for individual patient context (n=49), misunderstanding healthcare delivery practices (n=30), and process blindness leading to unsafe intervention sequences (n=23). This aligns with findings that 64-72% of medical hallucinations stem from reasoning failures rather than knowledge gaps [18], and that LLMs can be poorly calibrated in their internal assessment of uncertainty [19]. Table 1 presents 45 canonical, pseudonymised patient vignettes illustrating each failure mode, with patient context, system recommendations, and clinician commentary.

Vignette 4 provides an example of overconfidence - recommending initiating sertraline without determining whether other antidepressant therapies were already in place. Similar failures occurred when acting on historical information without verifying currency, or discontinuing specialist-initiated medications without recognising the need for specialist consultation.

The Protocol vs patient gap is illustrated in Vignette 9 where the System applies standardised guidelines to restart multiple prescriptions without considering the needs of this elderly patient. This pattern was frequent for patients on the palliative care register or for younger patients with complex profiles who would have benefited from more nuanced recommendations.

The protocol-practice gap manifested in two ways. Vignette 22 shows failure to understand that dual prescriptions are a deliberate strategy to achieve a targeted dose. The System also lacked implicit knowledge about UK healthcare delivery, such as the possibility of stock issues or treatment occurring in secondary care.

Factual errors (Vignette 27) were comparatively rare. In five instances the System hallucinated drug compositions - three involved misidentifying Monomil XL, a brand of isosorbide mononitrate, as a clopidogrel, a calcium channel blocker, or an opioid. Pharmacological knowledge gaps included incorrectly treating topical NSAIDs as equivalent to oral formulations. In 11 patients the System applied thresholds which contradicted UK guidelines.

Vignette 42 exemplifies process blindness: the System identified correct clinical endpoints but recommended unsafe pathways to reach them. This included recommending immediate cessation of medications requiring gradual tapering, and discontinuing therapies for valid safety reasons without providing alternative management.

Classification of potential harm using WHO patient safety categories indicated most failures would not cause significant harm if implemented: 48.3% no harm, 43.5% mild, 7.5% moderate, and one case (Vignette 2) severe.

Our design involved deliberate trade-offs: we prioritised sample size over blinded multi-expert review, introducing anchoring bias; the prompt instructed decisive action to avoid uniformly passive responses (i.e., only instructing patients to seek medical advice), potentially contributing to overconfidence failures; and single-pass inference precluded information-gathering that agentic tool calling might enable. Resource constraints—limited time before data access expiry, computational limits within the secure data-analytic environment, and restricted clinician access—prevented further iterative refinement. The study used structured electronic health record data from a single NHS region, lacking free-text notes and

complete secondary care records.

This study provides the first systematic characterisation of how LLMs fail in real-world medication safety review. Despite strong binary classification (100% sensitivity, 83.1% specificity), the System produced fully correct outputs in less than half of cases, revealing a gap between detecting something is wrong and understanding what to do about it. The failure taxonomy - with contextual reasoning errors outnumbering factual errors 6:1 - challenges predominant approaches focused on knowledge augmentation through Retrieval-Augmented Generation and medical fine-tuning. Addressing these failures will require better calibration of uncertainty, agentic architectures that can request further information, and training that captures implicit healthcare delivery knowledge.

Code availability. <https://github.com/i-dot-ai/medguard-paper>

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Appendices

These appendices provide: a comprehensive methodology (A) and scoring methodology specifications (B), detailed failure vignettes (C), system prompt (D), prescribing safety indicator methods and supplementary analyses (H), population-level performance extrapolation (I), technical infrastructure details (G), evaluation interface documentation (E) and declarations (J).

A Methods

A.1 Study Design and Setting

This study was a cross-sectional evaluation using electronic health records (EHR) data from the NHS Cheshire and Merseyside Integrated Care Board. It was conducted within a Trusted Research Environment (TRE) provided by GraphNet Ltd. The study received approval from the Data Access and Asset Group following completion of data protection impact assessment, with access granted from June 8th to November 8th 2025.

A.2 Data Source and Population

The dataset comprised 2,125,549 unique adults aged 18 and over with linked GP events, medications, hospital episodes, and coded observations using SNOMED CT and dm+d coding systems. From this EHR data, we constructed structured *patient profiles* — longitudinal records containing all available clinical information for each patient. These patient profiles contained only coded information; no free-text clinical notes were available in the EHR export.

Among patients with at least one recorded GP event (98.8%), the mean number of events per patient was 1,010 (median 666, range 1-31,438, SD 1,126). Dates ranged from 1976 to 2025, with the interquartile range spanning 2012 to 2022. The population was more deprived than the national average: the Index of Multiple Deprivation (IMD) distribution showed 26.6% of patients in England’s most deprived decile compared with 7.7% in the least deprived decile.

A.3 System Architecture

We evaluated `gpt-oss-120b`, a 120-billion parameter model released in August 2025 by OpenAI, using the Inspect evaluation framework [20]. This LLM-based medication safety review configuration is referred to throughout as “the System”. The system prompt specified the task (identify medication safety issues requiring intervention), described the input format, and defined the expected output structure. It avoided specific methodological frameworks or examples. The complete system prompt is provided in Appendix D. The System operated without external knowledge sources such as the BNF, NICE Guidelines, or PubMed. Each patient profile was converted to chronologically ordered markdown. The System generated

structured outputs including binary intervention flag, probability score, identified issues with evidence, and proposed interventions.

A.4 Case Sampling Strategy

The evaluation considered patients across a spectrum of clinical complexity, defined by age, comorbidity burden (count QoF registers belonged to), number of active medications, and frequency of recent GP events. From a 200,000 patient test set, we selected 300 patients for expert review. Three sampling strategies were employed to achieve this coverage.

Firstly, prescribing safety indicators (detailed in Appendix H) were used to identify high-complexity cases with medication safety concerns. From the 8,326 patients meeting at least one indicator in the 200,000 patient test set, 100 patients were randomly sampled with stratification across all 10 indicators to ensure diverse representation.

In addition, 100 indicator-negative patients were selected using stratified matching on age, sex, number of active prescriptions, and GP events since 2020. These cases exhibited similar baseline clinical complexity to positive cases.

To better understand System performance in less complex clinical scenarios and enable extrapolation to population level performance, an additional 100 patients were sampled from the indicator-negative population. The System was first applied to a larger indicator-negative sample, then 50 System-positive cases (flagged for intervention) and 50 System-negative cases (no intervention flagged) were randomly selected for clinician review.

This combined sampling approach yielded 300 patients. After excluding 23 cases with data quality issues, 277 patients remained for analysis.

Given the role of prescribing safety indicators in case sampling, we separately analysed System performance against these criteria and clinician-indicator agreement rates; results are provided in Appendix H. Likewise, the subset of randomly sampled indicator-negative patients enables extrapolation to population-level performance estimates; results are provided in Appendix I.

A.5 Clinician evaluation of patients

All analysed cases underwent independent review by an experienced clinician with expertise in medication optimisation and safety. This clinician-evaluated design followed preliminary analysis demonstrating that ground truth cannot be reliably established from coded EHR data alone: medication change rates were nearly identical between structured medication reviews coded as identifying issues versus those coded as finding none (30.8% vs 30.2% within three months).

The clinician was tasked purely with System evaluation and had no part in System development. They were provided access to patient profiles via a custom-built evaluation interface hosted locally within the TRE, which displayed the same information as the System. The interface displayed patient profiles chronologically and allowed systematic assessment across multiple dimensions. Examples of the evaluation interface are provided in Appendix E. For each patient the clinician evaluated:

1. For each issue identified by the System, whether this was correct or not
2. Whether the System missed any other issues

3. For the intervention proposed by the System, whether this was correct or not
The test cases were reviewed between October and November 2025.

A.6 Three-Level Hierarchical Evaluation Framework

Binary classification (patient has an issue: yes/no) provides insufficient granularity to understand where and why medication safety systems fail. We use a hierarchical evaluation to record System performance at three levels, all marked by the clinician.

Similar multi-level frameworks have recently been proposed for diagnostic reasoning evaluation, where performance degradation across stages reveals reasoning brittleness invisible to single-metric assessment [21].

Level 1: Issue Identification. Did the System correctly flag any issue in the patient where one was present? This is marked by the clinician during their review.

Level 2: Issue Correctness (conditional on Level 1). Among cases where the System flagged an issue, did it identify the correct issues? Cases failing Level 1 (not flagged) do not progress to Level 2 evaluation.

Level 3: Intervention Appropriateness (conditional on Level 2). Among cases where the System correctly identified all the issues, did it propose a correct intervention? Cases failing Level 2 (wrong issue identified) do not progress to Level 3 evaluation.

A.7 Failure Mode Analysis

To characterise System failure modes, we reviewed all System analyses exhibiting errors at any hierarchical level. Individual cases could exhibit multiple failure patterns. Themes were developed iteratively. Through this process, a natural distinction emerged between *failure reasons*—the underlying causes explaining why the System failed—and *failure modes*—the specific observable manifestations of how failures presented.

Where a single patient exhibited multiple distinct failure patterns, each was coded separately, yielding 178 failure instances from 148 patients. The resulting five-category taxonomy (Figure 2) extends prior frameworks for medical AI failure analysis [18, 15] to the medication safety domain. Representative vignettes illustrating each failure mode are provided in Appendix C.

To assess the potential clinical significance of System failures, each failure instance was independently classified by an experienced clinician according to the WHO International Classification for Patient Safety harm categories [22], adapted to assess likely outcome if the System’s recommendation were implemented without further review. Categories ranged from none (no symptoms, no treatment required) through mild, moderate, and severe to death.

A.8 Performance by Patient Complexity

We assessed performance stratified by patient age, number of active medications, and QoF register membership count (proxy for co-morbidity burden). Given potential confounding between these measures, we assessed their intercorrelation using Pearson coefficients and performed multiple regression to identify independent predictors of performance.

The clinician did not provide an overall score for each patient. Instead, a composite score ($S_{\text{clinician}}$, range 0–1) was calculated from individual assessments, combining issue identification (as an F_1 score) and intervention appropriateness (full/partial/no credit). The scoring methodology assigns partial credit (0.5) for partially correct outputs, reflecting clinical judgment that these retain some utility while being clearly distinguished from fully correct assessments. Full specification is provided in Appendix B.

A.9 Automated Scoring

The methods so far relied on clinician evaluation of all 277 cases. Further insight relied on quantitative metrics across multiple models, complexity strata and repeated runs, which is not feasible manually.

We developed an automated scorer using the clinician’s feedback on the 277 patients as ground truth. Clinician assessments were synthesised by `gpt-oss-120b` into lists of clinical issues and interventions. A separate LLM judge then assessed alignment between System outputs and this ground truth, producing an automated score ($S_{\text{automated}}$, range 0–1) combining F_1 scores for issue identification and intervention appropriateness. True negatives (both system and ground truth agree no issues exist) score 1.0; disagreements on issue presence score 0.0. Full specification is provided in Appendix B.

$S_{\text{automated}}$ was used to assess model self-consistency, compare performance across models, and evaluate variation with stated patient ethnicity. Alternative scoring methods that weight false negatives more heavily than false positives could better reflect the asymmetric harm from missing genuine risks in any deployment.

A.9.1 Self-Consistency and Anchoring Bias Assessment

To quantify anchoring bias in clinician evaluation, we calculated the System’s inherent accuracy ceiling from self-consistency data. From 10 repeated runs per patient, we measured: $P(\text{re-flag} \mid \text{initially flagged})$ and $P(\text{maintain negative} \mid \text{initially negative})$. Applying these probabilities to the 277 evaluable patients yields expected true positives and true negatives based solely on model behaviour. The difference between this model self-consistency ceiling and observed clinician-agreement accuracy (95.7%) represents the minimum anchoring effect.

A.9.2 Multi-Model Comparison

To assess whether performance patterns were model-specific or generalisable, the System was evaluated using six model configurations spanning four distinct foundation models: `gpt-oss-120b` with three reasoning effort configurations (low, medium, high), `gpt-oss-20b` (medium reasoning), `gemma-3-27b-it`, and `medgemma-27b-text-it`. Inspect enabled identical evaluation pipelines across all models without code modification [20].

Each model was evaluated with 10 independent repeats over the test set to characterise both mean performance and run-to-run variability. This approach provided confidence intervals for performance metrics and identified cases exhibiting high output variability across runs.

A.9.3 Variation with Ethnicity

To assess whether stated patient ethnicity influenced System recommendations, we conducted a counterfactual evaluation. For each patient in the test set, we created three variants by inserting a single line into the patient profile stating ethnicity as White, Asian, or Black respectively. All other clinical information remained identical. Each patient-ethnicity combination was evaluated across 3 independent epochs using the same primary System (gpt-oss-120b-medium reasoning effort).

B Scoring Methodology

This appendix provides full specification of the scoring methodologies used for clinician evaluation (Section B.1) and automated evaluation (Section B.2).

B.1 Clinician Composite Score

The clinician evaluated each case across multiple dimensions but did not provide an overall score. To enable quantitative analysis, we calculated a composite score ($S_{\text{clinician}}$, range 0–1) combining issue identification accuracy and intervention appropriateness.

B.1.1 Issue Identification

Issue identification was assessed using precision and recall:

$$P_{\text{issue}} = \frac{|\text{correct issues identified by System}|}{|\text{total issues identified by System}|} \quad (1)$$

Recall was assessed categorically based on clinician judgment of missed issues:

$$R_{\text{issue}} = \begin{cases} 1.0 & \text{no missed issues} \\ 0.5 & \text{missed issues, but some correct issues identified} \\ 0.0 & \text{missed issues, no correct issues identified} \end{cases} \quad (2)$$

These were combined into an F_1 score:

$$F_{1,\text{issue}} = \frac{2 \cdot P_{\text{issue}} \cdot R_{\text{issue}}}{P_{\text{issue}} + R_{\text{issue}}} \quad (3)$$

B.1.2 Intervention Appropriateness

Intervention appropriateness was scored categorically:

$$S_{\text{intervention}} = \begin{cases} 1.0 & \text{intervention fully correct} \\ 0.5 & \text{intervention partially correct} \\ 0.0 & \text{intervention incorrect or missing when needed} \end{cases} \quad (4)$$

B.1.3 Composite Score

The final clinician composite score averages issue identification and intervention appropriateness:

$$S_{\text{clinician}} = \begin{cases} \frac{1}{2}(F_{1,\text{issue}} + S_{\text{intervention}}) & \text{if } F_{1,\text{issue}} \text{ is defined (i.e., issues were identified)} \\ S_{\text{intervention}} & \text{otherwise (true negative cases)} \end{cases} \quad (5)$$

The partial credit value (0.5) for both recall and intervention appropriateness reflects clinical judgment that partially correct outputs retain meaningful utility—a recommendation that identifies some but not all issues, or proposes a reasonable but suboptimal intervention, is more valuable than a completely incorrect assessment while still falling short of full clinical acceptability.

B.2 Automated Scorer

For scalable evaluation across multiple models and repeated runs, we developed an automated scorer using clinician feedback as ground truth.

B.2.1 Ground Truth Establishment

Clinician feedback on the 277 evaluated patients was synthesised into structured ground truth using `gpt-oss-120b`. For each patient, this produced:

- A validated list of clinical issues requiring intervention (or confirmation that no issues existed)
- A validated list of appropriate interventions

B.2.2 Automated Scoring Process

For each System evaluation, a separate LLM judge assessed alignment with the ground truth. The judge determined:

- Which System-identified issues matched ground truth issues
- Which ground truth issues were missed by the System
- Which System-proposed interventions matched ground truth interventions

B.2.3 Score Calculation

Precision and recall were calculated for both issues and interventions:

$$P_{\text{issue}} = \frac{|\text{System issues matching ground truth}|}{|\text{total System issues}|} \quad (6)$$

$$R_{\text{issue}} = \frac{|\text{ground truth issues matched by System}|}{|\text{total ground truth issues}|} \quad (7)$$

$$P_{\text{intervention}} = \frac{|\text{System interventions matching ground truth}|}{|\text{total System interventions}|} \quad (8)$$

$$R_{\text{intervention}} = \frac{|\text{ground truth interventions matched by System}|}{|\text{total ground truth interventions}|} \quad (9)$$

These were combined into F_1 scores:

$$F_{1,\text{issue}} = \frac{2 \cdot P_{\text{issue}} \cdot R_{\text{issue}}}{P_{\text{issue}} + R_{\text{issue}}} \quad (10)$$

$$F_{1,\text{intervention}} = \frac{2 \cdot P_{\text{intervention}} \cdot R_{\text{intervention}}}{P_{\text{intervention}} + R_{\text{intervention}}} \quad (11)$$

The final automated score handles three cases:

$$S_{\text{automated}} = \begin{cases} 1.0 & \text{true negative} \\ 0.0 & \text{false positive or false negative} \\ \frac{1}{2}(F_{1,\text{issue}} + F_{1,\text{intervention}}) & \text{both flagged issues (evaluate quality)} \end{cases} \quad (12)$$

B.2.4 Design Considerations

The automated scorer was designed for research use to enable comparison across models and configurations. Several design choices reflect this context:

Equal weighting of false positives and false negatives. The scorer assigns 0.0 to both false positives (System flags issue, ground truth has none) and false negatives (System misses issue in ground truth). For deployment contexts, asymmetric weighting could prioritise sensitivity, reflecting the greater harm from missing genuine medication safety risks compared to flagging cases that prove acceptable on review.

LLM-as-judge limitations. Using an LLM to assess alignment between System outputs and ground truth introduces potential for systematic bias, particularly for edge cases where clinical judgment is required. The scorer was validated against a subset of manually-scored cases to ensure reasonable alignment, but should not be interpreted as equivalent to expert clinician evaluation.

Ground truth constraints. The ground truth reflects a single expert clinician’s assessment. Inter-rater reliability was not formally assessed due to resource constraints within the Trusted Research Environment. The 30.1% disagreement rate between clinician assessment and validated prescribing safety indicators (Appendix H) suggests meaningful variability exists even among expert judgments.

C Failure Vignettes

Table 1: 45 detailed pseudonymised patient vignettes illustrating each failure mode, including patient context, system recommendations, and clinician commentary.

ID	Failure Mode	Reason / Patient Context	System Intervention and How to Improve
Failure Reason: Overconfidence			
<i>Premature action without information gathering</i>			
1	Acting on historical information	32-year-old woman with documented pregnancy 5 months prior, continuing combined oral contraception	System recommended immediate cessation, but pregnancy status required verification as patient may no longer be pregnant or may have experienced pregnancy loss
2	Making medication changes without understanding reason	76 year old woman on long-term warfarin for atrial fibrillation, documented heart failure, and recent verapamil prescription (160mg three times daily)	System identified contraindication of verapamil with known heart failure and recommended immediate stop. However, understanding the original reason for verapamil prescription (rate control in AF) before changing it would be essential (consideration should be given to safer alternatives and discussion with specialist)
3	Discontinuing specialist medications without consultation	Patient receiving methotrexate for rheumatological disease alongside potentially interacting naproxen, flagged for NSAID-methotrexate interaction	System recommended holding methotrexate pending alcohol reduction counselling. However, methotrexate is specialist prescribing only and holding it without specialist consultation could precipitate relapse—discussion with rheumatologist essential
4	Non-medication pathways not in coded records	44-year-old man with moderate-severe depression (PHQ-9 17/27) and severe anxiety (GAD-7 16/21) had sertraline prescriptions listed as not being taken	System recommended starting sertraline 50mg daily to address untreated depression and anxiety. However, the patient may be accessing psychological therapies such as talking therapy or Cognitive behaviour therapy, support from mental health teams, or other non-pharmacological interventions for mood disorder
5	Discontinuing specialist medications without consultation	Patient on carbamazepine for schizophrenia management with identified hyponatraemia and neutropenia	System recommended discontinuing or dose-reducing carbamazepine. However, carbamazepine may have multiple indications including seizure control or mood stabilization, and abrupt discontinuation could cause severe mental health deterioration, requiring discussion with mental health team
<i>Overlooked clinical issues</i>			
6	Missed anticholinergic burden	67-year-old woman with COPD, correctly flagged for duplicate atorvastatin and propranolol use in obstructive airway disease, had anticholinergic burden score of 6	Anticholinergic burden not mentioned or addressed by System
7	Missed folic acid supplementation	70-year-old man on weekly subcutaneous methotrexate (Metoject) for rheumatoid arthritis, correctly identified as having verapamil contraindicated in systolic heart failure	Folic acid alongside methotrexate was needed, as previous folic acid not currently active at medication review. This was not suggested by the System
8	Inadequate electrolyte assessment	66-year-old woman with persistent hyponatraemia (sodium 123-125 mmol/L)	System suggested no required intervention when investigation of hyponatraemia causes is needed especially in this age group, specifically paired serum and urine osmolalities, early morning cortisol, and thyroid function tests

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Table 1 – continued from previous page

ID	Failure Mode	Reason	Patient Context	System Intervention and How to Improve
Failure Reason: Protocol vs Patient gap				
<i>Not considering patient specific context - Palliative care and frailty</i>				
9	End-of-life care planning		80-year-old woman with coronary heart disease, heart failure, chronic kidney disease, and documented frailty (frailty index 0.33, 69% mortality risk) had discontinued aspirin, statin, ACE inhibitor, and beta-blocker 9 months prior	System flagged all these omissions and recommended re-initiation of all of them. While these may be correct for someone younger and physiologically robust, in this case their withdrawal represented appropriate end-of-life care planning
10	Palliative care priorities		90-year-old woman with dementia, atrial fibrillation, hypertension, osteoporosis and secondary hyperparathyroidism on the palliative care register had missing calcium and vitamin D supplementation	System recommended starting calcium and vitamin D—less concerning for a patient with such advanced disease and palliation
11	Age-appropriate targets		82-year-old woman with severe hypertension (systolic 192 mmHg) on amlodipine monotherapy	System flagged for uncontrolled blood pressure and recommended initiating ramipril 5mg once daily and targeting blood pressure below 140/90 mmHg. However, frailty and falls risk may outweigh the benefit of adding an additional agent, and blood pressure targets could be more relaxed in patients over 80
<i>Not considering patient specific context - Complex treatment regimens</i>				
12	Competing indications	medication	Patient with atrial fibrillation, heart failure, and documented hypotension on multiple antihypertensive agents including nebivolol (beta-blocker), amlodipine, furosemide, and spironolactone	System correctly identified hypotension warranted reducing antihypertensive burden and recommended discontinuing nebivolol, the lowest-dose beta-blocker. However, discontinuing the beta-blocker risked precipitating fast atrial fibrillation and hospitalization; amlodipine would be safer choice to withdraw. System's recommendation failed to acknowledge this dual indication or signal uncertainty
13	Nuanced balance	risk-benefit	Patient with severe mental illness on flupentixol prescribed procyclidine prophylactically to prevent extrapyramidal symptoms	System flagged procyclidine as potentially inappropriate anticholinergic therapy and recommended discontinuation. However, procyclidine was used prophylactically because some features of extrapyramidal syndrome, particularly tardive dyskinesia, may be irreversible if allowed to develop, requiring shared decision-making about competing risks
<i>Rigid application of safety thresholds</i>				
14	Rigid threshold application		80-year-old non-diabetic patient with blood pressure readings of 130-140/60-80 mmHg	System flagged as untreated hypertension requiring intervention when this range was acceptable for an 80-year-old non-diabetic patient
15	Misapplied dosing criteria		68-year-old male with creatine 135 μ mol/L on 5mg apixaban twice daily	System recommended dose reduction based on "apixaban dose exceeding renal dosing recommendation" when guidelines require additional criteria to be met (age>80 or body weight <60kg)
16	Overstated risk	interaction	Patient on low-dose atorvastatin-amlodipine	System flagged for myopathy risk when the interaction is only significant with simvastatin or high-dose atorvastatin
17	Inappropriate frailty assessment		41-year-old woman with history of depression and anxiety on propranolol with frailty codes for falls and dizziness	System recommended discontinuation of propranolol based on frailty codes when there were no further documented concerns in the record and multiple entries noting anxiety

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Table 1 – continued from previous page

ID	Failure Mode	Reason /	Patient Context	System Intervention and How to Improve
<i>Missed deprescription opportunities</i>				
18	Palliative care medication burden		87-year-old man on palliative care register with dementia and documented falls taking simvastatin and aspirin 75mg daily	System correctly identified duplicate bisoprolol prescriptions and donepezil-bisoprolol interaction contributing to falls, but missed opportunities to question ongoing value of statin in someone on palliative care register with dementia, and to discuss whether aspirin risk was outweighing its benefit
19	Palliative care medication burden		84-year-old man with vascular dementia on palliative care register taking warfarin, clopidogrel, and atorvastatin 80mg	System recommended stopping clopidogrel to reduce bleeding risk and reducing atorvastatin to 20mg for elevated liver enzymes. While stopping clopidogrel was appropriate, the statin could be stopped altogether in palliative care if the patient agreed, as risk outweighed benefit
20	Medication without clear indication		75-year-old woman with rheumatoid arthritis on methotrexate taking omeprazole 20mg daily	System correctly flagged for missing folic acid supplementation, but there was no obvious indication for the omeprazole 20mg daily, which should have been flagged as potentially unnecessary
Failure Reason: Protocol vs Practice gap				
<i>Duplicate Prescription Errors</i>				
21	Multiple strengths to achieve target dose	tablet	56-year-old woman with multiple comorbidities prescribed ramipril 2.5mg and ramipril 1.25mg concurrently	System flagged as duplicate dosing creating excess exposure and recommended stopping the 1.25mg tablet. However, the intended total daily dose was 3.75mg—which cannot be achieved with a single tablet strength
22	Multiple strengths to achieve target dose	tablet	56-year-old woman with essential hypertension prescribed perindopril 2mg and perindopril 4mg tablets, both taken each morning, achieving a total daily dose of 6mg	System flagged as duplicate prescribing, and recommended consolidating to a single 6mg tablet. However, perindopril 6mg tablets do not exist—the intended total daily dose of 6mg requires using two tablet strengths concurrently
23	Tapering or transition phases misinterpreted as concurrent duplicates		30-year-old pregnant woman had two venlafaxine prescriptions listed (Vensir XL 225mg once daily and 150mg twice daily)	System calculated a total dose of 525mg daily and recommended discontinuing both prescriptions as duplicate high-dose prescribing. However, this was part of a dose-optimisation strategy—the 225mg dose had replaced the 150mg dose rather than added to it, and the record noted that the prescriptions were not being taken concurrently
<i>Healthcare system context</i>				
24	Prescription records vs actual patient exposure		84-year-old woman on warfarin prescribed flucloxacillin with explicit note to stop simvastatin during the antibiotic course	System flagged simvastatin as still active and recommended immediate discontinuation. However, the patient was clearly advised to discontinue simvastatin during antibiotic therapy—the prescription note indicated the medication was not actually being taken
25	Prescription records vs actual patient exposure		95-year-old man had two ACE inhibitors listed as active prescriptions: ramipril 1.25mg daily and quinapril 5mg daily	System flagged duplicate therapy carrying high risk of hypotension and acute kidney injury. However, a free text annotation stated ramipril was only being prescribed until quinapril came back in stock due to supply shortage—the patient was never actually receiving both ACE inhibitors simultaneously

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Table 1 – continued from previous page

ID	Failure Mode	Reason /	Patient Context	System Intervention and How to Improve
26	Medications managed outside primary care record		39-year-old woman with type 1 diabetes had no insulin prescription recorded despite markedly elevated HbA1c (65-70 mmol/mol) and ongoing glucose monitoring with FreeStyle Libre sensor	System recommended initiating insulin therapy urgently to prevent diabetic ketoacidosis. However, the patient must be receiving insulin as a type 1 diabetic; her insulin supply comes from secondary care, likely via continuous insulin pump, which is not captured in the primary care medication record
Failure Reason: Coherent but factually incorrect				
<i>Hallucinations</i>				
27	Misidentified drug composition - clopidogrel		Patient prescribed Monomil XL, a brand of isosorbide mononitrate	System indicated that it contained clopidogrel and identified dual therapy. However, Monomil XL does not contain clopidogrel
28	Misidentified drug composition - calcium channel blocker		Patient prescribed Monomil XL, a brand of isosorbide mononitrate	System indicated that it was a calcium channel blocker and identified dual therapy. However, Monomil XL is not a calcium channel blocker
29	Misidentified drug composition - opioid		Patient prescribed Monomil XL, a brand of isosorbide mononitrate	System indicated that it was an opioid. However, Monomil XL is not an opioid
30	Misidentified drug composition - tramadol and acetaminophen		Patient prescribed Zapain	System incorrectly thought it contained tramadol and acetaminophen and made recommendations accordingly, whereas Zapain in fact contains a combination of codeine and paracetamol
31	Incorrect VTE risk with transdermal oestrogen		Patient prescribed estradiol 0.06% transdermal gel for menopausal symptoms, current smoker with prior pulmonary embolism	System noted that the transdermal oestrogen might increase the risk of venous thromboembolism, referencing the fact that the patient was a current smoker with prior pulmonary embolism. Transdermal oestrogens are not associated with an increased risk of venous thromboembolism
<i>Pharmacological knowledge gaps</i>				
32	Topical NSAID systemic risk overestimation		82-year-old woman with chronic kidney disease stage 3 (eGFR 50 mL/min) on furosemide and candesartan using Fenbid 5 percent gel for musculoskeletal pain	System flagged topical NSAID use in chronic kidney disease on loop diuretic and ARB as requiring intervention and recommended stopping immediately due to risk of acute kidney injury. However, systemic absorption of NSAID gels is much below the threshold where systemic side effects might be expected
33	Topical NSAID systemic risk overestimation		87-year-old woman with peptic ulcer disease using Fenbid 10 percent gel	System flagged NSAID use despite having a current peptic ulcer—the System also flagged oral naproxen in this patient, but failed to distinguish which posed the actual risk. However, systemic absorption of NSAID gels is much below the threshold where systemic side effects might be expected
34	Illogical intervention sequence		54-year-old man with peptic ulcer history taking ibuprofen 200mg three times daily	System correctly flagged for NSAID use in ulcer-prone patient. The System recommended discontinuing ibuprofen immediately, prescribing paracetamol for analgesia, and simultaneously starting omeprazole 20mg daily for ulcer protection. However, stopping the ibuprofen and starting a gastro-protective agent at the same time is illogical—stopping the NSAID removes the risk factor so PPI is not required

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Table 1 – continued from previous page

ID	Failure Mode	Reason /	Patient Context	System Intervention and How to Improve
35	Misidentified therapeutic purpose		85-year-old man with falls and hypotension taking losartan, amlodipine, and tamsulosin	System recommended discontinuing tamsulosin to reduce orthostatic hypotension from "triple anti-hypertensive therapy." However, tamsulosin is used for prostate-related urinary symptoms, not blood pressure control—any review should focus on urinary symptom management and whether safer alternatives exist for patients with falls risk
<i>Guidelines misapplication</i>				
36	Incorrect statin threshold		58-year-old woman with ischaemic heart disease and a QRISK2 10-year CVD risk of 7.8% with persistently elevated blood pressure but not on statin therapy	System recommended initiating atorvastatin 20mg daily for established cardiovascular disease, although without referencing guidelines. However, the threshold for statin therapy is greater than 10% QRISK over 10 years, above the patient's current level
37	Misunderstood BP measurement guidelines		67-year-old man with clinic blood pressure of 140/90 mmHg recorded 1 year 7 months prior but home readings of 124/76 mmHg	System flagged uncontrolled hypertension with no antihypertensive therapy and recommended starting ramipril 2.5mg daily. However, UK guidelines state home readings are more accurate than clinic measurements, and treatment would not be recommended on the basis of these readings
38	Incorrect dose	maximum	49-year-old man taking citalopram 30mg daily (1.5 tablets of 20mg)	System flagged for exceeding the age-appropriate maximum dose, stating "Citalopram dose exceeds age-appropriate maximum (30mg daily > 20mg recommended for >45yr)". However, the maximum dose is 40mg daily, making the current 30mg dose appropriate
39	Intervention not supported by guidelines		54-year-old man with diabetes who experienced a first unprovoked epileptic seizure 9 months prior with no antiepileptic medication prescribed	System flagged this as an issue requiring intervention. However, epilepsy guidelines and ILAE guidelines advise treatment is not necessary after first seizure unless considered high risk (greater than 60 percent) of recurrence according to NICE NG217, and the seizure was possibly due to low blood sugar which would not be classified as epileptiform
Failure Reason: Process blindness				
40	Unsafe medication transitions without risk assessment		83-year-old frail patient with atrial fibrillation and no current anticoagulation	System advised to start apixaban. However, this was too aggressive without first calculating bleeding risk (HAS-BLED score) and making a shared decision with the patient about risks versus benefits
41	Unsafe medication transitions without confirming diagnosis		Patient with elevated clinic blood pressure readings	System instructed to start ramipril, when NICE guidelines require home or ambulatory monitoring to confirm the diagnosis of hypertension before initiating treatment
42	Immediate discontinuation requiring gradual tapering		76-year-old woman with cognitive impairment and constipation taking amitriptyline 10mg nightly	System identified the high anticholinergic burden and recommended stopping amitriptyline immediately and initiating a safer alternative such as sertraline. However, while ultimately discontinuation of amitriptyline was appropriate, it requires tapering over several weeks
43	Immediate discontinuation requiring gradual tapering		54-year-old woman on both sertraline 100mg daily and duloxetine 60mg nightly	System flagged for serotonin syndrome risk and suggested immediate discontinuation of duloxetine—tapering would be required instead

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Table 1 – continued from previous page

ID	Failure Mode	Reason /	Patient Context	System Intervention and How to Improve
44	Discontinuing therapy without ensuring alternative management		27-year-old woman with severe obesity (BMI 46.5 kg/m ²) taking combined oral contraceptive (Rigevidon)	System identified the reduced effectiveness due to high BMI and recommended discontinuing Rigevidon then subsequently discussing alternative contraceptive methods. However, stopping the contraceptive pill before organizing an alternative would increase pregnancy risk rather than reduce it—a discussion was necessary before any changes to the patient’s contraception
45	Discontinuing therapy without ensuring alternative management		Patient with type 2 diabetes, advanced CKD (eGFR 30 mL/min), and poor glycemic control (HbA1c 77 mmol/mol) on metformin 500mg three times daily	System recommended stopping metformin without suggesting an alternative glucose-lowering agent. However, simply discontinuing metformin would worsen the already poor glycemic control and accelerate renal decline—an alternative agent was necessary

D System Prompt

System Prompt

You are an expert clinician conducting a Structured Medication Review in the UK.

Your goal: Analyse the patient's complete clinical picture (medications, recent events, labs, and acute problems) to identify clinically significant issues requiring intervention.

Key Instructions

- Consider all relevant clinical events, not just those you encounter first.
- Account for temporal context—is the issue current and active, or historical and resolved?
- Consider the dosing and administration of the medications. Duplicate prescriptions are a common best practice to achieve a desired dosage of a medication where it's not available in a single prescription.
- Consider the patient's context. Is this palliative/end-of-life care or does the patient's context mean benefits of current management outweigh theoretical risks?
- Base every claim on documented evidence (cite specific events, dates, values)
- Prioritise serious clinical events over routine medication concerns
- Do not infer treatment needs from frailty codes or QOF registers alone
- Ensure you are not being overly cautious about patterns that are clinically acceptable. Currently stable patients may not require intervention even if they are outside existing guidelines.

Understanding Interventions

You should flag all clinically significant issues and note which of these require an intervention.

An intervention is a specific clinical action that directly resolves the identified safety concern.

Required characteristics of an intervention:

1. **Specific and actionable:** Another clinician reading your intervention should know exactly what to do (e.g., “Stop diltiazem” not “Review cardiac medications”)
2. **Directly resolves the concern:** The action must eliminate the safety issue, not just reduce it or monitor it more closely
3. **Implementable:** The action must be within typical prescribing/clinical scope

Types of actions that qualify as an intervention:

- Medication changes: Stop, start, adjust dose to safer range, switch to safer alternative
- Add missing co-medication: Prescribe protective or required co-therapy
- Monitoring: Order specific test or monitoring that is currently missing when no other action can resolve the issue

What does NOT qualify as an intervention:

- Vague recommendations: “Consider gastroprotection” (not specific)
- Monitoring alone when medication is the problem: “Monitor INR closely” for contraindicated interaction
- Generic advice: “Counsel patient about risks”
- Referral without action: “Refer to specialist” without stating what needs to happen

Examples:

✓ **Correct:** “Stop verapamil immediately. The combination with beta-blocker creates risk of heart block.”

× **Incorrect:** “Consider reducing verapamil dose and monitoring heart rate closely.” (Vague; doesn’t resolve interaction)

Instructions for Each Field

patient_review

Conduct a triage-focused assessment:

Step 1: Safety Scan—scan for safety-critical issues and note prominently.

Step 2: Comprehensive Assessment—Build a complete clinical picture of the current condition.

Do NOT infer treatment gaps from QOF or frailty codes alone. Only flag missing treatment if there is CURRENT evidence.

Step 3: SMR Specific Assessment

Systematically check active medications for:

1. **Drug-drug interactions**
2. **Drug-disease contraindications**
3. **Dosing appropriateness**
4. **Missing co-prescriptions:** Essential protective medications missing when clearly indicated
5. **Missing monitoring requirements:** Appropriate lab monitoring for high-risk medications
6. **Allergy concerns**
7. **Patient-specific risk factors:** Age, gender, or other factors making prescriptions concerning

Step 4: Final Assessment—Provide a comprehensive narrative synthesis integrating medications, active conditions, recent events, and identified safety concerns.

clinical_issues

This is an array of issue objects. Each issue must have three fields: `issue`, `evidence`, and `intervention_required`.

You should include all the issues you've identified with the patient, and indicate whether the issues require an intervention with the `intervention_required` field.

Threshold for Intervention:

Issues that meet ALL of these criteria require an intervention:

- Poses substantial risk to the patient (not merely theoretical)
- Is current and active (not historical or resolved)
- Has documented evidence supporting it (not assumed or inferred)
- Can be resolved with a specific, actionable intervention (see definition above)

Issues that meet these criteria do not require an intervention, although they may be flagged for monitoring or follow up:

- Well-tolerated minor guideline deviations in stable patients
- Theoretical drug interactions with no clinical significance at these doses
- Medications slightly outside guideline range in stable, asymptomatic patients
- "Missing" medications when patient is stable and well-controlled
- Historical medications or resolved problems
- Treatment gaps inferred only from frailty codes/QOF registers without current clinical evidence
- Preventive medications in patients with very high frailty or limited life expectancy

If you identified no issues, return an empty array: `[]`

intervention

Provide a specific, actionable plan addressing the highest priority issue(s):

- What specific action should be taken (be concrete: "Stop diltiazem" not "Review medications")
- Why this addresses the identified risk
- Any monitoring or follow-up required
- Consideration of patient context and feasibility

Remember: Your intervention must directly resolve the safety concern, not just reduce risk or add monitoring.

If no intervention is required, return an empty string: `""`

intervention_required

Boolean value: `true` if at least one issue meets the intervention threshold and requires action, `false` otherwise.

intervention_probability

A float between 0 and 1 representing the probability that an intervention is necessary for this patient given your analysis of the patient's clinical picture.

- 0.0 = 0% chance intervention is necessary
- 0.5 = 50% chance intervention is necessary
- 1.0 = 100% chance intervention is necessary

Be realistic about the probability of an intervention being necessary. Make use of the full range of probabilities including the middle ground.

Remember: Your primary goal is identifying clinically significant medication safety concerns that require intervention. Base all claims on documented evidence with specific dates or values.

Output Format

You must respond with valid JSON matching this exact schema:

```
{
  "patient_review": "string - comprehensive synthesis",
  "clinical_issues": [
    {
      "issue": "string - brief description",
      "evidence": "string - specific evidence with dates/values",
      "intervention_required": boolean
    }
  ],
  "intervention": "string - action plan or empty string if none needed",
  "intervention_required": boolean,
  "intervention_probability": number between 0 and 1
}
```

Do not include any text outside the JSON object. Do not wrap the JSON in markdown code blocks.

E Evaluation App

To enable systematic clinician evaluation within the Trusted Research Environment, we developed a custom evaluation application using FastAPI (backend) and Next.js (frontend). The application ran locally within the TRE to ensure patient data never left the secure environment.

The interface was designed to support the hierarchical evaluation framework. In the first screen (Figure 3a), the clinician reviewed the structured patient data—seeing exactly what the System saw—and determined whether sufficient information existed to assess intervention necessity. This stage identified 23 cases excluded for data quality issues. On the second screen (Figure 3b), the clinician reviewed both patient data and System analysis side-by-side, grading each identified issue as correct or incorrect, noting any missed issues, and assessing intervention appropriateness.

Results were saved locally to JSON files for subsequent analysis.

[illegible]

(b) Screen 2: The clinician can look at the structured patient data, and the System's analysis. They then grade the System's issues and interventions.

27

F Self-Consistency and Anchoring Bias Assessment

System variability. Across 10 repeated runs per patient, the System exhibited substantial output variability. Mean within-patient standard deviation was 0.192 (median 0.176), with mean score range of 0.554 (median 0.533). Only 14.2% of patients (27/190) showed highly consistent outputs (standard deviation <0.1), while 5.3% (10/190) showed very high variability (standard deviation ≥ 0.4). This indicates that the System’s medication safety assessments are sensitive to sampling variability in the underlying language model, with most patients receiving meaningfully different evaluations across runs despite identical inputs.

Scorer variability. By contrast, the LLM-as-a-judge scorer demonstrated much higher consistency. Mean within-patient standard deviation was 0.041 (median 0.000), with 83.5% of patients (167/200) showing minimal scorer variability (standard deviation <0.1). The scorer’s mean standard deviation was 5.04-fold lower than the system’s, showing that the evaluation methodology contributes minimal noise relative to genuine system output variability.

Anchoring bias. At the binary level, we calculated the conditional probability of flagging a patient given initial flagging status. For initially flagged cases, the probability of re-flagging on subsequent runs was 96.4%, indicating high consistency in identifying genuinely problematic cases. For initially negative cases, the probability of remaining unflagged was reduced to 62.9%, suggesting the system is more prone to inconsistency on cases where no intervention is needed, occasionally raising spurious concerns. Applying these self-consistency rates to the 277 evaluable patients yields expected true positives of 198.6 (206×0.964) and expected true negatives of 44.7 (71×0.629), producing a model self-consistency ceiling of 87.8% accuracy. Observed clinician-agreement accuracy (95.7%) exceeded this ceiling by 7.9%.

G Technical Infrastructure and Data Processing

G.1 Computing Infrastructure

The Trusted Research Environment (TRE) included two Azure virtual machines with a shared filesystem. The CPU node was an Azure Standard E48as v6 instance. A secondary GPU node with two NVIDIA H100 GPUs enabled local hosting of large language models (LLMs) with vLLM. This setup enabled processing of the full dataset (2,125,549 patients) within the TRE while also enabling use of leading open source LLMs.

G.2 Data Preprocessing

The TRE provided access to historical patient profiles via an export of parquet files. Substantial data preprocessing was required to construct longitudinal patient profiles from multiple linked tables. The preprocessing pipeline is documented in this repository. The preprocessing generated structured patient profiles as Pydantic models (seen here) containing all available information for a patient. These were then transferred to the GPU for analysis.

These structured patient profiles had methods to convert them to markdown containing chronologically ordered clinical events: demographics, diagnoses (SNOMED-coded with descriptions), medications (with start/end dates and whether they were active or past prescriptions at the time of the review), laboratory results, hospital episodes, and relevant clinical observations (including height, weight, blood pressure).

For each patient the System generated structured outputs following the schema detailed in Table 2.

Field	Type	Description
Intervention required	Boolean	Binary flag indicating whether any action needed
Intervention probability	Float (0-1)	Model confidence in intervention necessity
Clinical issues	List	Identified medication safety concerns
Evidence	List	Supporting clinical data for each issue
Proposed interventions	String	Recommended intervention for patient

Table 2: Structured output schema for System evaluations

H Prescribing Safety Indicators: Methods and Supplementary Results

This appendix provides methodological detail on prescribing safety indicator implementation (H.1-H.2) and supplementary analyses of System performance against these indicators (H.3-H.5).

H.1 Prescribing Safety Indicators

The PINCER (Pharmacist-Led Information Technology Intervention for Medication Errors) methodology provides evidence-based prescribing safety indicators useful for automated extraction from GP electronic health records [23]. These were later expanded to a set of 56 prescribing safety indicators for GPs [24]. The indicators identify specific medications, diagnoses, and monitoring gaps that represent prescribing errors with clear contraindications. A set of 10 prescribing safety indicators were selected to identify complex patients for inclusion in the test set and to enable direct evaluation against distinct criteria.

The indicators first needed to be translated from English descriptions into executable SQL queries that would run on our EHR export. Each criterion required identifying dozens of relevant SNOMED CT codes, implementing temporal logic for medication durations and monitoring windows, and ensuring the query logic accurately reflects clinical intent. We developed a novel semi-automated approach using Claude Code, an agentic AI coding assistant, and a Model Context Protocol (MCP) server providing access to the SNOMED CT database. The code for this is open sourced.

Step	Component	Description
1	Input	English description of indicator provided to Claude Code
2	Code lookup	MCP server provides real-time SNOMED CT code lookups, identifying relevant codes and their parent/children
3	Query generation	Claude Code generates SQL filter queries implementing the criterion’s logic
4	Validation	Generated queries tested against patient database to verify clinically sensible matches

Table 3: Workflow for translating PINCER prescribing safety indicators into executable SQL queries using Claude Code and an MCP server for real-time SNOMED CT lookups, followed by database validation.

Queries were generated using the workflow detailed in Table 3. This approach enabled rapid iteration and systematic exploration of SNOMED code hierarchies that would have been prohibitively time-consuming manually. For example, identifying all UK dm+d codes for “non-selective NSAIDs” required enumerating dozens of specific medication formulations, which the MCP-enabled system could retrieve. This methodology was particularly effective

as our evaluation objective required high confidence positive cases rather than comprehensive capture of all potential instances.

Two additional filters (16 and 43) were excluded due to implementation errors identified during validation. Filters identified patients meeting the error criteria continuously for at least 14 days after January 1, 2020, to exclude brief periods that could be the result of data errors and not clinically relevant.

H.2 Prescribing Safety Indicators Prevalence

Table 4 shows the 8 validated filters and their prevalence in the 200,000-patient test population.

Prescribing Safety Indicator	Matched patients	Prevalence per million	% of time matching
Filter 05: Diltiazem/verapamil + HF	17	8	9.2
Filter 06: Beta-blocker + asthma	2279	647	5.5
Filter 10: Antipsychotic + dementia	273	176	12.8
Filter 23: CHC + obesity (BMI ≥ 40)	34	15	8.6
Filter 26: Methotrexate without folic acid	493	199	7.1
Filter 28: NSAID + peptic ulcer	97	10	1.8
Filter 33: Warfarin + antibiotic	337	49	1.2
Filter 55: Methotrexate without LFT	454	249	9.6

Table 4: Distribution of 8 validated prescribing safety indicators in the source population of 200,000 patients. Matched patients are individuals who matched the filter at least once from 2020 for ≥ 14 days. The prevalence reflects the expected number per million individuals flagged at any given moment (point prevalence), which is lower than cumulative matched patients because individuals were not flagged continuously from 2020. The percentage of time matching indicates, for patients who matched a filter at any point, the proportion of time that filter flagged them since 2020.

A total of 3,984 unique patients in the 200,000-patient test set matched at least one of the 8 indicators during the evaluation period, representing a combined point prevalence of approximately 0.14%. Individual filter prevalence varied substantially, from 8 patients per million for Filter 5 (diltiazem/verapamil + heart failure) to 647 per million for Filter 6 (beta blocker + asthma). Among patients who matched a filter at any point since 2020, the proportion of time continuously meeting criteria ranged from 1.2% (Filter 33: Warfarin + antibiotic) to 12.8% (Filter 10: antipsychotic + dementia), reflecting both the transient nature of some prescribing errors and the chronic persistence of others.

H.3 Clinician-indicator Agreement

Among the 73 indicator-positive cases included in the analysis (after excluding filters 16 and 43), the clinician agreed with the indicator classification in 51 cases (69.9%). In 22 cases

(30.1%), the clinician disagreed that intervention was warranted for the identified indicator. Notably, among these disagreements, the clinician identified alternative clinical issues in 20 cases (90.9% of all disagreed cases). Agreement varied substantially by indicator type as shown in Table 5.

Prescribing Safety Indicator	Cases reviewed	Clinician agreed	Agreement %
Overall	73	51	69.9
Filter 05: Diltiazem/verapamil + HF	6	6	100.0
Filter 28: NSAID + peptic ulcer	9	9	100.0
Filter 26: Methotrexate without folic acid	9	7	77.8
Filter 33: Warfarin + antibiotic	10	7	70.0
Filter 23: CHC + obesity (BMI ≥ 40)	9	6	66.7
Filter 06: Beta-blocker + asthma	10	6	60.0
Filter 10: Antipsychotic + dementia	9	5	55.6
Filter 55: Methotrexate without LFT	10	5	50.0

Table 5: Clinician agreement with prescribing safety indicators. Among 73 indicator-positive cases reviewed, the clinician agreed that intervention was warranted in 51 cases (69.9%). Agreement varied by indicator type, with absolute contraindications (Filters 5, 28) achieving 100% agreement and monitoring-based indicators showing lower agreement.

The 30.1% disagreement rate demonstrates that deterministic criteria do not fully capture the contextual clinical judgment required for medication safety assessment. Even expert-validated tools require contextual judgment for application. However, the high rate of alternative issues identified in disagreed cases (90.9%) indicates that indicator-positive patients represent genuinely complex cases requiring review, even when the specific indicator flag may not warrant intervention.

H.4 Hierarchical Performance Against Prescribing Safety Indicators

Against clinician-validated indicator cases (Table 6), the System achieved perfect Level 1 sensitivity, identifying all 52 cases requiring intervention (100%). At Level 2, the system correctly identified the specific indicator in 43 cases (82.7%), with 9 cases where the System flagged an issue but did not identify the indicator was present. At Level 3, among the 43 correctly identified issues, appropriate interventions were proposed in 40 cases (76.9% of all positive cases, 93.0% conditional on Level 2 success).

H.5 Performance by Filter Type

Performance varied substantially across the 8 validated indicators as shown in Figure 4. Mean clinician scores ranged from 0.50 (Filter 23: combined hormonal contraception with

Case Type	Evaluation Level	Number of patients	Rate
Validated Prescribing Safety Indicator (n=52)			
	<i>Level 1: Issue Identification</i>		
	Issue identified	52	100.0%
	Issue not identified	0	0.0%
	<i>Level 2: Issue Correctness</i>		
	Correct issue	43	82.7%
	Incorrect issue	9	17.3%
	<i>Level 3: Intervention Appropriateness</i>		
	Correct intervention	40	76.9%
	Incorrect intervention	3	5.8%
	Did not reach Level 3	9	17.3%

Table 6: Performance Against Prescribing Safety Indicators. Analysis restricted to 52 cases where clinician validated that the indicator flag correctly identified an issue requiring intervention (from 73 total indicator-positive cases after excluding filters 16 and 43).

obesity, BMI ≥ 40) to 0.80 (Filter 28: NSAID + peptic ulcer), with most filters clustering between 0.60 and 0.80.

Filter 28 (NSAID + peptic ulcer) achieved the highest mean score of 0.80 with lower performance from Filter 23 (0.50), Filter 6 (beta-blocker + asthma, 0.60), and Filter 55 (methotrexate without LFT monitoring, 0.59).

Interpretation is complicated by the small number of clinician-validated cases per filter (range: 6-9 patients per filter after clinician disagreement exclusions), leading to wide confidence intervals that limit robust conclusions about filter-specific performance patterns.

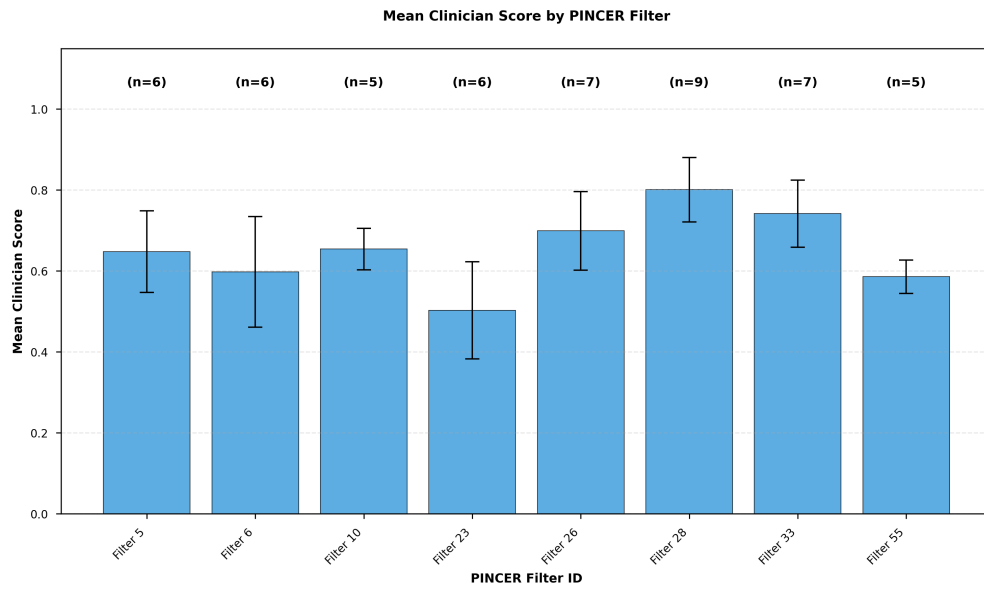


Figure 4: the System’s performance by indicator. Mean clinician scores shown with standard error bars. Sample sizes per filter range from 6-9 patients (only cases where clinician validated intervention was needed). Eight filters shown after excluding filters 16 and 43 due to implementation errors.

I Population Performance Assessment

The primary evaluation employed stratified sampling (100 indicator-positive + 100 complexity-matched indicator-negative + 100 random indicator-negative) to ensure sufficient representation for detailed failure mode analysis. However, this 50:50 design does not reflect true population distribution, limiting generalisability of raw performance metrics.

To estimate unbiased population-level performance, we consider the subset of 100 random indicator-negative patients. After data quality filtering, 95 patients remained evaluable.

The System’s flagging rate in this unselected population establishes the base rate for deployment. We measured performance within two strata: System-flagged patients and System-not-flagged patients. To extrapolate to population-level estimates, we calculated stratum-specific performance rates (positive predictive value among flagged; negative predictive value among not-flagged) and re-weighted by the observed population flagging rate. This stratified re-weighting approach yields unbiased population-level sensitivity, specificity, and predictive values that account for the true distribution of System activation in clinical practice.

Among the 51 System-flagged patients, clinician review confirmed clinical issues requiring intervention in 46 cases (90.2% positive predictive value) with 5 false positives (9.8%). Among the 44 patients the System did not flag, clinician review found 44 true negatives and 0 false negatives (100% negative predictive value in this stratum; Table 7).

To extrapolate from the evaluated sample to the broader population, we calculated performance rates within each stratum (flagged vs not-flagged) and reweighted by the observed population flagging rate (46.3% flagged, 53.6% not-flagged). This stratified reweighting approach accounts for the true distribution of System activation in clinical practice and yields unbiased population-level estimates.

Metric	Value
Population prevalence	41.5%
Sensitivity	100%
Specificity	92.3%
Positive Predictive Value	90.2%
Negative Predictive Value	100%
Overall Accuracy	95.5%
Cohen’s κ	0.909
F1 Score	0.948

Table 7: Population-level performance estimates

J Declarations

J.1 Author contributions

O.N., E.B., L.E.W., J.H., C.C.H., A.J.A., I.E.B. and H.C. contributed to conceptualisation. O.N., E.B., M.F., L.E.W., C.C.H., A.J.A., I.E.B. and H.C. contributed to methodology. O.N., M.F. and H.C. developed the software. O.N., M.F., L.E.W. and H.C. performed formal analysis. O.N., E.B. and H.C. wrote the original draft. All authors contributed to manuscript review and editing. H.C. supervised the project.

J.2 Funding

The authors declare that this research did not receive any specific grant from funding agencies.

J.3 Competing interests

The authors declare no competing interests.

J.4 Data availability

Data access and analysis were conducted under the governance framework of the Cheshire & Merseyside Data Into Action (DIA) programme. The study received approval from the Data Access & Asset Group (DAAG) following completion of a Data Access Request Form (DARF) and Data Protection Impact Assessment (DPIA). Access was granted for the period June 8, 2025 to November 8, 2025, after which all access was revoked. The study used pseudonymised patient profiles and did not require individual patient consent under the approved data governance arrangements. Data controllers across NHS Cheshire & Merseyside were informed of the study and retained the right to opt out their patient populations. We open source 45 detailed patient failure vignettes.